Rachel R. Sharp

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EDUCATION

PhD University of North Carolina at Chapel Hill (UNC) | Neuroscience 2021 – 2026
Dissertation: *Genetic and Social Risk for Bipolar Disorder: Equity Gaps, Clinical Readiness, and the Responsible Use of Polygenic Scores*
Certificate in Bioinformatics and Computational Biology
NSF Graduate Research Fellow

BS University of Oklahoma (OU) | Psychology, Criminology Minor 2016 – 2019
Summa Cum Laude (4.0 GPA)

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant | University of North Carolina at Chapel Hill 2021 – 2026
Principal Investigator: Dr. Michael Love (rotation 2021; joined lab 2022)

Project: The Effects of Social Risk on Genetic Liability for Bipolar Disorder

- Designed and implemented a population-scale study of genetic and social risk for bipolar disorder in 115,275 participants of the *All of Us* Research Program, integrating whole-genome sequencing, electronic health records, and participant surveys
- Found that social risk contributed considerable explanatory power beyond genetic risk alone, with high social burden conferring substantial risk even among individuals in the lowest deciles of polygenic liability
- Advocated for holistic models of psychiatric risk that integrate genetic liability with social and environmental context, showing that precision psychiatry frameworks that omit social risk provide an incomplete picture of individual-level risk
- Coordinated collaboration among clinicians, geneticists, biostatisticians, and population health researchers at UNC and Duke through to publication (first author, Human Genetics and Genomics Advances, 2026)

Project: Statistical Validation of Genetic Variants Associated with Neuropsychiatric Disorders in Genome-wide Association Studies (GWAS)

- Computationally identified genetic variants associated with neuropsychiatric disorders from GWAS and Massively Parallel Reporter Assay (MPRA) datasets
- Integrated results with Quantitative Trait Loci datasets to interrogate genetic mechanisms contributing to disease penetrance

Project: Statistical Pipeline Development for Standardized Analysis of MPRA Data Across Institutions within the International Impact of Genomic Variation on Function Consortium (IGVF)

- Developed streamlined pipelines to systematically and reproducibly analyze MPRA data across five internationally funded centers
- Organized teams across centers to develop discrete pipeline portions and led cross-center meetings, setting analytical standards for a multi-institutional consortium

Bioinformatic Analyst | OU Health Sciences Center Core Facilities 2020 – 2021
Principal Investigator: Dr. Kenneth Jones

- Independently produced custom scripts in R, Python, Bash, and other languages to process and analyze data from single cell and bulk sequencing projects
- Collaborated with 20+ PIs on diverse projects including metastasis of glioblastomas, cancer subtypes in zebrafish models, cell morphology shifts in myopia, and genetic signatures of type 2 diabetes
- Produced high-quality figures and descriptions for papers and grants
- Contributed analyses to 6 co-authored papers, including PLOS Pathogens, Endocrinology, and Frontiers in Immunology

Lab Manager | OU Visual Neuroscience Lab 2018 – 2019
Principal Investigator: Dr. Michael Wenger

- Managed and directed 20+ undergraduate research assistants
- Conducted studies examining the effects of iron deficiency on cognition in college-aged women, women undergoing smoking cessation, and perimenopausal women
- Collected and analyzed EEG data from EGI nets and gel-based caps
- Trained two graduate students and 20+ undergraduate research assistants on EEG net application and safety, data acquisition, and data processing

PEER-REVIEWED PUBLICATIONS GOOGLE SCHOLAR

1. **Sharp, R. R.**, Hysong, M. R., Mealer, R. G., Raffield, L. M., Glover, L., & Love, M. I. (2026). Considering social risk alongside genetic risk for bipolar disorder in the All of Us Research Program. *Human Genetics and Genomics Advances*, *7*(4), 100665. https://doi.org/10.1016/j.xhgg.2026.100665
2. Bruxel, E. M., Rovaris, D. L., Belangero, S. I., . . . **Sharp, R. R.**, . . . Montalvo-Ortiz, J. L. (2025). Psychiatric genetics in the diverse landscape of Latin American populations. *Nature Genetics*, *57*(5), 1074–1088. https://doi.org/10.1038/s41588-025-02127-z
3. IGVF Consortium. (2024). Deciphering the impact of genomic variation on function. *Nature*, *633*(8028), 47–57. https://doi.org/10.1038/s41586-024-07510-0
4. Yashchenko, A., Bland, S. J., Song, C. J., Ahmed, U. K. B., **Sharp, R.**, . . . Zimmerman, K. A. (2023). Cx3cr1 controls kidney resident macrophage heterogeneity. *Frontiers in Immunology*, *14*, 1082078. https://doi.org/10.3389/fimmu.2023.1082078
5. Larabee, J. L., Doyle, D. A., Ahmed, U. K. B., Shadid, T. M., **Sharp, R. R.**, Jones, K. L., Kim, Y. M., Li, S., & Ballard, J. D. (2023). Discovery of Hippo signaling as a regulator of CSPG4 expression and as a therapeutic target for *Clostridioides difficile* disease. *PLOS Pathogens*, *19*(3), e1011272. https://doi.org/10.1371/journal.ppat.1011272
6. Nelson, B. N., Daugherty, C. S., **Sharp, R. R.**, Booth, J. L., Patel, V. I., Metcalf, J. P., Jones, K. L., & Wozniak, K. L. (2022). Protective interaction of human phagocytic APC subsets with *Cryptococcus neoformans* induces genes associated with metabolism and antigen presentation. *Frontiers in Immunology*, *13*, 1054477. https://doi.org/10.3389/fimmu.2022.1054477

7. De Jong, N. P., Rudolph, M. C., Jackman, M. R., **Sharp, R. R.**, Jones, K., Houck, J., Pan, Z., Reusch, J. E. B., MacLean, P. S., Bessesen, D. H., & Bergouignan, A. (2022). Short-term adaptations in skeletal muscle mitochondrial oxidative capacity and metabolic pathways to breaking up sedentary behaviors in overweight or obese adults. *Nutrients*, *14*(3), 454. https://doi.org/10.3390/nu14030454

8. Scalzo, R. L., Foright, R. M., Hull, S. E., Knaub, L. A., Johnson-Murguia, S., Kinanee, F., Kaplan, J., Houck, J. A., Johnson, G., **Sharp, R. R.**, Gillen, A. E., Jones, K. L., Zhang, A. M. Y., Johnson, J. D., MacLean, P. S., Reusch, J. E. B., Wright-Hobart, S., & Wellberg, E. A. (2021). Breast cancer endocrine therapy promotes weight gain with distinct adipose tissue effects in lean and obese female mice. *Endocrinology*, *162*(11), bqab174. https://doi.org/10.1210/endocr/bqab174

9. Matye, D. J., Wang, H., Luo, W., **Sharp, R. R.**, Chen, C., Gu, L., Jones, K. L., Ding, W.-X., Friedman, J. E., & Li, T. (2021). Combined ASBT inhibitor and FGF15 treatment improves therapeutic efficacy in experimental nonalcoholic steatohepatitis. *Cellular and Molecular Gastroenterology and Hepatology*, *12*(3), 1001–1019. https://doi.org/10.1016/j.jcmgh.2021.04.013

CONFERENCE PRESENTATIONS AND ABSTRACTS

1. **Sharp, R. R.**, Hysong, M. R., Mealer, R. G., Raffield, L. M., Glover, L., & Love, M. I. (2026, January 28–29). *More than the sum of our genes: How social risk shapes the development of bipolar disorder* [Poster presentation]. **Festival of Genomics & Biodata**, London, United Kingdom.

2. **Sharp, R. R.**, Glover, L., Raffield, L. M., & Love, M. I. (2024, November 5–9). *Assessing the impact of social exposomes on genetic liability for bipolar disorder* [Poster presentation]. **American Society of Human Genetics Annual Meeting**, Denver, CO, United States.

3. **Sharp, R. R.**, Glover, L., Raffield, L. M., & Love, M. I. (2024, September). *Assessing the impact of social exposomes on genetic liability for bipolar disorder* [Oral presentation]. **UNC Department of Genetics Annual Retreat**, Myrtle Beach, SC, United States.

4. **Sharp, R. R.**, McAfee, J. C., Lee, S., Rosen, J., Love, M. I., & Won, H. (2023, May). *Statistical validation of GWAS-identified neuropsychiatric variants* [Poster presentation]. **Genomics of Brain Disorders Conference**, Wellcome Genome Campus, Hinxton, United Kingdom.

5. **Sharp, R. R.**, Broadaway, K. A., McAfee, J. C., Mohlke, K. L., Love, M. I., & Won, H. (2022, September). *Computing MPRA and QTL variant overlap for schizophrenia-associated, fine-mapped variants* [Lightning talk and poster presentation]. **Impact of Genomic Variation on Function Consortium Annual Meeting**, Bethesda, MD, United States. **Best Poster Presentation Award (2nd place).**

6. Wenger, M. J., **Sharp, R. R.**, McCollum, A., De Stefano, L., Rhoten, S., & Worth, T. (2019, May 17–22). *Negative impacts of iron deficiency on visual category learning quantified in terms of dopaminergic status and brain energy expenditure* [Poster presentation]. **Vision Sciences Society Annual Meeting**, St. Pete Beach, FL, United States. Abstract published in *Journal of Vision*, *19*(10), 244c. https://doi.org/10.1167/19.10.244c

7. **Sharp, R. R.**, Foster, S., & Carvallo, M. (2017, May). *The fundamental attribution error and prejudice reversal* [Poster presentation]. **University of Oklahoma Psychology Conference**, Norman, OK, United States. **Best Undergraduate Research Poster Award.**

TEACHING EXPERIENCE

Volunteer R Instructor | UNC School of Medicine 2022 – 2024

- Taught the R programming language and general coding practices to 40 students enrolled in the How to Learn to Code R course
- Taught students how to load and work with packages, debug code, use APIs, and organize their scripts

Writing Assistant | OU Honors College 2018 – 2019

- Taught advanced written language skills to 25+ students in various Honors College writing courses which aim to develop critical thinking and a broad worldview
- Assisted students in developing tools for persuasion, clarity, and emotion in writing

Peer Learning Assistant | OU University College 2018 – 2019

- Tutored 50+ students in Intro Biology, Java I and II, Research Methods, Intro Psychology, and Study Skills
- Worked individually and with teams of tutors to facilitate a student-led learning experience based on Socratic questioning
- National Tutor Certification

Ceramics Instructor | San Miguel Middle School 2016

- Independently developed a ceramics program for an underprivileged middle school in Tulsa, Oklahoma
- Fundraised and purchased all necessary supplies for teaching and implementing a ceramic class
- Wrote lesson plans and independently instructed twelve students on the basics of ceramics and 3D art at their middle school over the course of a semester
- Seeded the school with program details, supplies, and additional funds to continue running the program

LEADERSHIP, MENTORSHIP, AND SCIENCE OUTREACH

Neurosciences Student Executive Committee | President 2022 – 2025

Represented 100+ graduate students in regular meetings with program directors and administrative staff, gathering feedback from students and implementing it as program changes. Overhauled the qualifying exam, rewrote the Annual Progress Review reports, established anonymous reporting channels and first-year check-ins, and built community through the annual retreat and events such as the chili cook-off

Human Functional Regulatory Genomics Group Organizer 2023 – 2025

Planned, organized, recruited speakers for, and facilitated monthly trainee-focused meetings with presentations and discussions on topics in the field of functional genomics

ASHG Session Moderator and Abstract Reviewer 2024

Reviewed and scored submitted abstracts for the annual American Society of Human Genetics conference. Led and moderated the conference session “Phenomenal PheWAS”

Culbreth Middle School STEAM Day Volunteer Instructor 2024

Led a presentation to rotating classes of middle schoolers on DNA, graduate education, and career paths in science

WinSPIRE Student Recruitment Team 2022 – 2024
Recruited high school students from the research triangle area to participate in WinSPIRE (Women in Science Promoting Inclusion in Research Experiences), a summer mentorship program aimed at creating a space where students from minoritized genders can gain confidence in their ability to pursue a career in STEM by doing real scientific research with the help of early-career scientists. Mentored the students during the duration of the program and led workshops on science communication to a broad audience

Peer Mentor 2022 – 2024
Mentored first-year graduate students to guide them through rotations, classes, and adjusting to life in graduate school

Letters to a Pre-Scientist 2022 – 2024
Exchanged letters with a middle-school pen pal from an underprivileged school to humanize STEM professionals, demystify STEM careers, and inspire future exploration in STEM

BBSP In-Person Visits Planning Committee 2022 – 2025
Worked with the graduate admissions office to plan and organize events for incoming graduate students

UNC Summer Undergraduate Research Experience Mentor Summer 2023
Mentored undergraduate students participating in a summer research internship to guide them through the process of choosing a post-undergraduate career path, applying to graduate schools, and finding their place in academia

Triangle Brain Bee 2022 – 2023
Wrote questions for, judged, and organized tutoring sessions for the Brain Bee, an annual spelling-bee style competition engaging middle school students with topics in neuroscience

Honors Mentorship Program Director 2018 – 2019
Independently expanded and directed the program, assembled an executive team, organized events, secured funding sources, and provided mentoring opportunities for Honors College freshmen

SCIENCE WRITING AND COMMUNICATION EXPERIENCE

Author | *American Scientist*, Macroscope 2025
- Interviewed Dr. Oliver Bear Don't Walk IV about building intersectionality into medical informatics and wrote "Finding Humanity in Health Data" for a general science readership
- Developed the piece through ComSciCon-Triangle science communication training and took it through the magazine's editorial process to publication

Contributing Author | *BrainPost* 2023 – 2025
- Transformed peer-reviewed neuroscience publications from top-tier journals into accessible summaries for non-specialized audiences
- Covered diverse topics from neuroplasticity to brain-computer interfaces, balancing scientific accuracy with engaging readability

Writer & Editor | *The Pipettepen* 2022 – 2025

- Authored original articles on scientific topics of personal interest
- Edited submissions from fellow graduate students, providing feedback to enhance clarity, narrative flow, and accessibility

Blog Scheduler, Translation Coordinator, and Editor | *NC DNA Day Blog* 2022 – 2023

- Broadened blog accessibility by developing and overseeing Spanish translation program, increasing translated content from 3 articles over two years, to 17 articles in a single semester
- Managed editorial calendar and coordinated communication between writers, editors, and translators

SELECTED WRITING PORTFOLIO

"Finding Humanity in Health Data" | *American Scientist* 2025

- Interview-driven commentary exploring the integration of social context into health data systems

"When Practice Makes Perfect: The Neurons Behind Skilled Movement" | *The Pipettepen* 2024

- Interview-based profile of ongoing research on NDNF neurons and their role in motor learning

"The Need for Diversity in Large-Scale Genetic Studies" | *The Pipettepen* 2023

- Analytical commentary on demographic representation gaps in genomics and implications for health equity

"Neuroscience-Backed Strategies to Help You Learn More Effectively" | *BrainPost* 2024

- Synthesis of recent cognitive neuroscience research into practical learning strategies

GRANTS, HONORS, AWARDS, AND SCHOLARSHIPS

Featured Author, Inside HGGA Interview Series 2026
Invited by *Human Genetics and Genomics Advances* (ASHG) for its author interview series following first-author publication; interview forthcoming on ashg.org

National Science Foundation Graduate Research Fellowship Program 2023
Research grant received covering $111,000 in stipend over 3 years, in addition to funds covering all costs for tuition, fees, insurance, and travel

Genomics of Brain Disorders Conference Bursary Travel Awardee 2023

Best Poster Presentation Award (2nd place) 2022
Awarded at the annual Impact of Genomic Variation on Function Consortium meeting as a second-year graduate student in competition with post-docs from internationally funded centers

UNC Department of Neuroscience T32 2022

Big Woman on Campus Award 2019
Awarded to individuals who have excelled in the areas of scholarship, character, leadership, and service to the OU community

Regent's Award for Outstanding Juniors 2019

Dean's Award for Outstanding Honors College Leadership 2019

John Benjamin Rotary Scholarship 2019
For high-achieving students and children of Tulsa first responders

President's Award for Outstanding Sophomores 2018

Ruby Brakebill Scholarship 2018
For students with a strong social conscience and leadership abilities

Patti Johnson Wilson Scholarship 2017
For high-achieving students in the liberal arts who also maintain employment

President's International Travel Fellowship 2017

National Merit Scholar 2016

PROFESSIONAL DEVELOPMENT & MEMBERSHIPS

ComSciCon Triangle Workshop Attendee 2025

Science Communicators of North Carolina Society Member 2024 – 2025

COMMUNITY OUTREACH

Community Organizer 2024 – 2026
Worked with local community-support organizations to maintain food distribution networks, table at rallies, and engage the public in discussions about supporting the local community

Crisis Text Line 2018 – 2019
Certified Crisis Counselor volunteer providing de-escalation services to individuals in crisis

Oklahoma Department of Corrections 2017
Volunteered with the OK Messages Project connecting incarcerated parents to their children

Honors at Oxford Study Abroad Program Summer 2017

TECHNICAL SKILLS

Programming and tools: R, Python, Bash, Unix, GitHub